BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, DI KHAN

INTERMEDIATE (11TH CLASS) EXAMINATION, 2022

PHYSICS

Past Paper 2022 • Paper Code: DIK-HSSC-XI-2022(A)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A

Time Allowed: 20 min | **Maximum Marks:** 18 | **Paper Code:** DIK-HSSC-XI-2022(A)

Q1. Choose the correct option i.e. A,B,C,D and fill it up in the correct bubble in the given MCQs sheet.

Qi. Which of the following is dimension less variable?

(A) Decay constant
(B) Plank's constant
(C) π Pi
(D) Refractive index

Qii. The percentage uncertainty in the volume of a cube is 3%. The percentage uncertainty in one side of the cube is_____

(A) 3%
(B) 1/3%
(C) 1%
(D) 1.5%

Qiii. If $\Delta = -3i + 3j$,the vector is lying in which quadrant?

(A) I
(B) II
(C) III
(D) IV

Qiv. Which one is correct for a body to be in complete equilibrium?

(A) $\Sigma FX = 0$
(B) $\Sigma\tau = 0$
(C) $\Sigma Fy = 0$
(D) $\Sigma F = 0\ \Sigma r = 0$

Qv. The quantities which are constant in elastic collision are ___

(A) K.E
(B) Momentum
(C) K.E and Momentum
(D) Angular Momentum

Qvi. The range and maximum height of projectile are equal for angle of projection of _____

(A) 30°
(B) 45°
(C) 76°
(D) 90°

Qvii. The efficiency of diesel engine is:

(A) 25-30%
(B) 35-40%
(C) 45-50%
(D) 55-60%

Qviii. The atmosphere is held to earth by _____

(A) Wind
(B) Gravitty
(C) Clouds
(D) Rotation of earth

Qix. The rate of change of angular momentum is_____

(A) Force
(B) Impulse
(C) Work
(D) Torque

Qx. 1 Pascal second = _____

(A) 10 poise
(B) 100 poise
(C) 10^3 poise
(D) 10^6 poise

Qxi. According to equation of continuity AV = constant, this constant is equal to _____

(A) Mass
(B) Volume flow rate
(C) Energy density
(D) Viscosity

Qxii. Tuning of radio set on example of _____

(A) Mechanical resonance
(B) Musical resonance
(C) Electrical resonance
(D) Free vibration

Qxiii. Two springs of constants K_1 K_2 are joined in series. The effective spring constant of the combination is given by_____

(A) $(K_1+K_2)/2$
(B) $K_1 + K_2$
(C) $(K_1K_2)/(K_1+K_2)$
(D) K_1K_2

Qxiv. The equation for the speed of longitudinal waves is _____

(A) √(δ/E)
(B) √(Tx1/m)
(C) [illegible in source scan]
(D) √(E/δ)

Qxv. The principal of young's double slit experiment is based on the _____

(A) Amplitude
(B) Frequency
(C) Velocity
(D) Wavelength

Qxvi. The tip of needle does not give a sharp, it is due to _____

(A) Polarization
(B) Refraction
(C) Diffraction
(D) Interference

Qxvii. The S.I unit of entropy is _____

(A) J.K
(B) JK^{-1}
(C) J.mol
(D) $J.mol.K^{-1}$

Qxviii. The difference between Newton's calculated value of speed of sound in air and in experimental value was _____

(A) 16%
(B) 60%
(C) 10%
(D) 5%

SECTION – B

Time Allowed: 2:40 Hrs | **Maximum Marks:** 67

Q2. (Marks 40) Answer any TEN questions of the following. All questions carry equal marks: [40]

i. How many radians account for circumference of a circle? How many staradians accounts for circumference of a sphere?

ii. Explain the warning "Never use a large wrench to tighten a small bolt".

iii. Find equation for maximum height of a projectile.

iv. A machine needs 1000J of energy to raise a 10Kg block at a distance of 6m. What is the efficiency?

v. Why is rifle barrel rifled?

vi. Why do the golf balls have dimple?

vii. What is damped oscillation? Give example.

viii. Differentiate the transverse and longitudinal waves motion.

ix. Discuss the effect of pressure on speed of sound in a medium.

x. Can we apply Huygens's principle to radar waves?

xi. Is it possible according to 2nd law of thermodynamics to construct a heat engine heat is free from thermal pollution.

xii. What is least count error? How can least count error to reduced?

xiii. Define static equilibrium with example.

SECTION – C

Maximum Marks: 27

Note: Note: Attempt any Three Questions. All questions carry equal marks:

Q.3 a. Calculate the value of escape velocity of a body and show that it is equal $11.2 Kms^{-1}$

b. What should be the orbital speed to launch a satellite in circular orbit 900km above the surface of earth?

Note: [Q.4 and Q.5 not present in source scan — ilmkidunya hosts a single-page subjective scan ending at Q.3(b)]